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AI-BASED EMERGENCY VEHICLE PRIORITIZATION

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Abstract - This paper details the design and implementation of an integrated Artificial Intelligence (AI) system for real-time emergency vehicle (EV) prioritization within urban traffic networks, utilizing the Simulation of Urban Mobility (SUMO) and its real-time Application Programming Interface, TraCI. The core problem addressed is the critical delay EVs—such as ambulances and fire trucks experience at controlled intersections, which severely compromises response times. The proposed solution combines the high-performance object detection capabilities of YOLOv8 with the dynamic simulation and control environment provided by SUMO and TraCI. The system employs a client-server architecture where the AI client processes vehicle detections, computes optimal signal changes, and executes instantaneous path clearance commands via TraCI.

Index Terms - Artificial intelligence, Emergency vehicle prioritization, Intelligent transportation systems, Traffic signal control, YOLOv8, SUMO, TraCI.

I. INTRODUCTION

In modern cities, urban mobility has become massive logistical puzzle but its flaws become a matter of life and death when they get in the way of emergency services. The traditional signals may lack the "big picture" awareness needed to clear a path through a complex grid, leading to dangerous delays and unpredictable response times. We can't rely on static infrastructure to handle the dynamic, high-staked nature of a life-saving mission.

For solving this I have developed a system that would move away from reactive signaling and towards intelligent, real-time intervention. By using a client-server setup which is an external AI "brain" that would keep a constant eye on where an emergency vehicle is and what the surrounding traffic looks like. We can achieve this by tapping into the Traffic Control Interface (TraCI) within the SUMO simulation environment. This will create a live loop where the AI can "talk" to the city's traffic grid and clearing a green-wave path multiple intersections ahead of the vehicle before it even arrives.

The real breakthrough here is about how we've managed to balance speed with the city-wide stability. By establishing a low-latency feedback loop between the AI and the road, we've proven that we could significantly slash emergency response times without causing any gridlock. More than just a faster ambulance, it's about creating a smarter and more adaptive urban environment where technology works behind the scenes to save lives while still keeping the city moving.

II. EASE OF USE

The overall practicality and accelerated development cycle of the emergency vehicle prioritization system stem directly from the inherent ease of use of its three most foundational components: YOLOv8, SUMO, and TraCI.

1) YOLOv8 (Emergency Vehicle Detection)

YOLOv8 will offer a significant enhancement in usability compared to the previous object detection models, simplifying the AI integration process:

Simple Workflow: The design offers a simple workflow with ease for all the stages, including data preparation.

- Integration with Python: It directly integrates with the standard environments for the Python programming language and has a clean and high-level API, which is referred to as Ultralytics. This makes it easy to install and develop the model through iterations for training, validation, and testing processes.
- Deployment Readiness: YOLOv8 offers readiness for a wide range of deployment models.

2) SUMO (Simulation Environment)

SUMO (Simulation of Urban MObility) will help in developing and testing the complex traffic control strategies:

- Open-Source Nature: Its open-source license eliminates cost and licensing restrictions, making it accessible on a global level.
- Real-World Network Import: SUMO's functionality allows for easy import of real-world road networks. This enables researchers to quickly set up complex, realistic simulation scenarios without intensive manual configuration required by proprietary tools.
- Microscopic Detail: The detailed microscopic simulation provides a clear quantifiable understanding on how the AI intervention affects individual vehicles.

3) TraCI (Real-Time Control Interface)

TraCI (Traffic Control Interface) acts as a crucial link that ensures the AI's logic to be easily translated into control commands:

- Intuitive Python API: TraCI exposes its functionality through a simple and intuitive Python API. This design will make both querying real-time traffic data and sending control commands straightforward.
- Focusing on Logic, Not Protocol: The simplicity of the API reduces the complexity of integrating external AI logic to great extent that allows developers to focus primarily on refining the decision-making algorithms instead of managing low-level network communication protocols.
- Reliable Execution: The command-response structure of the TraCI API helps in reliable and rapid execution of the time-critical commands like instantaneous traffic light phase changes using(setPhase).

III. LITERATURE REVIEW

This review confirms that saving lives with emergency vehicles requires a total shift in thinking. Old-school methods like Genetic Algorithms or Reinforcement Learning are great but works only for general flow and are

too slow and rigid for the chaotic nature of an approaching ambulance.

To get the sensing right, I chose YOLOv8 which is a major step-up from earlier versions. It hits the sub-10ms latency mark that we need, which means this system can "see" and identify an emergency vehicle even in a cluttered environment without any dangerous lags.

Talking about the virtual testing ground, SUMO beats the other simulators because it lets us zoom in on the "micro" details. The real game-changer is the TraCI API. While most simulators are built for big-picture city planning, TraCI gives us a direct line to the traffic lights. It allows the AI to step in and flip the signal to green at the exact moment it's needed, providing the kind of precision and real-time control that standard planning tools just cannot match.

IV. SYSTEM OVERVIEW

A. Architecture and Components

1) Real-Time Sensing (The AI Input)

This component is responsible for acquiring precise timely data about the emergency vehicle (EV).

- Technology: YOLOv8 Object Detection Architecture.
- Function: YOLOv8 is what you might call "the eyes of the system," and works exactly like the camera on the side of the road or the on-board sensor. The end result is that the system has a very efficient model—its "anchor-free" model and "decoupled head"—that keeps the accuracy very high and the lag under 10 ms. The lag is very important here since that gives the AI "lead time" to get through the intersection before the emergency vehicle arrives.
- Technology: SUMO (Simulation of Urban MObility) and AI Core Logic.
- Function: SUMO provides the indispensable microscopic simulation environment (the "virtual city") where individual vehicle behaviors and stops are accurately modeled. The AI Core Logic (implemented as the TraCI client) takes the YOLOv8 data and, considering the SUMO environment's current state (queues, signal phases), computes the optimal intervention. This involves identifying the minimal set of traffic signal groups that must receive a green phase to clear the EV's path while strategically minimizing the disruption to general traffic.

2) Execution and Closed-Loop Control (The TraCI Interface)

This component ensures the computed decision is applied instantly and that the system remains responsive.

- Technology: TraCI (Traffic Control Interface).

- Function: TraCI is the communication layer that enables the AI logic to control the SUMO environment instantaneously. It operates on a client-server architecture, managing a synchronized interaction loop. The client uses TraCI's simple Python API to:
 - Retrieve live data (position, signal states).
 - Execute atomic commands like setPhase to change the traffic signal state or setRoute to clear feeder lanes.

This execution creates the instantaneous "green wave" required for uninterrupted passage. The cyclical, synchronous nature of the TraCI interaction ensures that the AI is constantly monitoring and adapting to the EV's real-time movement until the priority period is terminated.

times of emergency services. This in-depth analysis will engage with the subject matter, which will particularly involve the lifesaving need of Emergency Vehicle Prioritization (EVP) and the complex algorithms that accommodate intelligent systems.

B. Comparative Analysis of Core Technologies

Intelligent Traffic Management Systems (ITMS) signify the onset of a radical shift in the planning, management, and maintenance of urban transport infrastructure. This has been set in motion by the collective evolution of computational, sensing, and communication technologies. Essentially, the advent of ITMS has been in response to the growing problem of urban traffic congestion, which can be considered one of the most substantial factors in economic losses, air pollution, and the delayed response

Component	Selected Technology	Key Advantage	Alternatives / Previous Methods	Limitations of Alternatives
Detection Engine	YOLOv8	Optimal real-time performance with high accuracy (mAP) and <10 ms inference latency on embedded hardware; uses simplified anchor-free architecture.	Earlier YOLO versions (v3, v4), R-CNN models	Slower inference not suitable for sub-10 ms control; lower accuracy/speed balance; complex anchor-box configuration.
Simulation Platform	SUMO	Integrates TraCI for instantaneous external control; microscopic vehicle modeling (stops, delays, queues); open-source (EPL 2.0) with no licensing restrictions.	MATSim, CityFlow, Commercial Simulators	Limited or no TraCI-like APIs; designed mainly for large-scale or agent-based planning, not real-time atomic control; high licensing cost.
Control Interface	TraCI	Enables synchronized, low-latency two-way communication for real-time commands (e.g., atomic setPhase); modular and easy-to-use Python API for AI integration.	Simple File I/O, Proprietary APIs (COM interfaces)	High communication delay; no guaranteed synchronized state updates; difficult low-level protocol management; limited direct signal control.

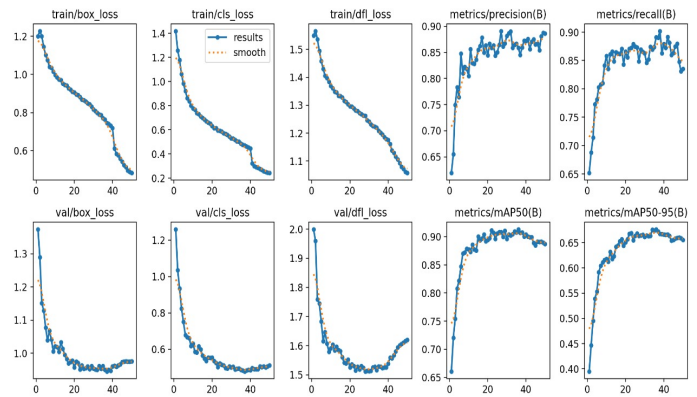
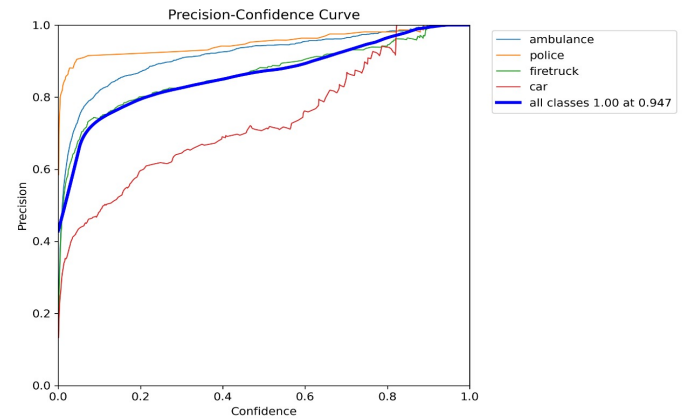
C. Operational Steps

The process is like a continuous feedback system, with its AI logic and simulation environment being constantly integrated so that a way is cleared for the passage of the emergency vehicle. Here is the process:

Phase 1: Startup & Sensing Everything begins by initiating the SUMO system start, loading the map and traffic data, accompanied by the TraCI connection for the AI brain simulation interface. After the system has been initialized, the YOLOv8 model serves as the eyes for the entire system, looking out for the emergency responders. Because the system has a latency rate below 10 ms, it can immediately detect the ambulance, check the speed, and monitor the location before conveying the information directly to the core system logic.

Phase 2: Logic and Precise Control Once an emergency vehicle has been recognized, "look ahead" logic, courtesy of TraCI, allows the AI to evaluate upcoming intersections for signal phases and traffic jams ahead of time. It does far more than turn a light green; instead, a "green wave" time, calculated from the ambulance' arrival time, is taken into account. By employing the "setPhase" command, the AI manages to let traffic signals switch at precise times to facilitate a smooth journey for the ambulance with delays to a mere minimum across intersecting traffic flows.

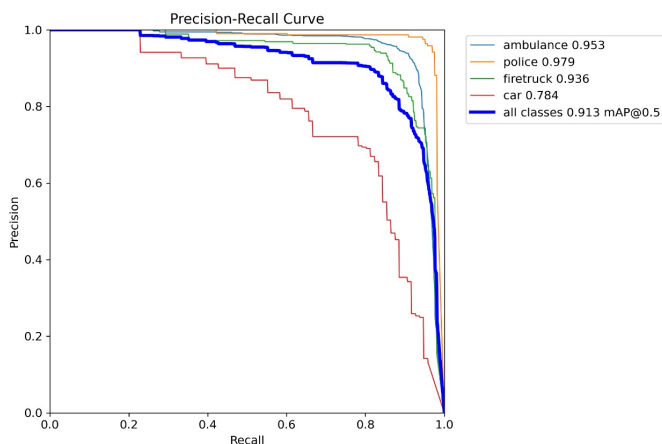
Phase 3: Coordination & Reset Once the system is locked onto the vehicle, it continues in an endless loop, helping it through every intersection that it passes during its route until the situation is alleviated. Once the vehicle passes through the last intersection before continuing on its route, the AI communicates through TraCI that the traffic flow should be returned to normal. This allows the network to return to its normal flow as soon as the priority route is passed.

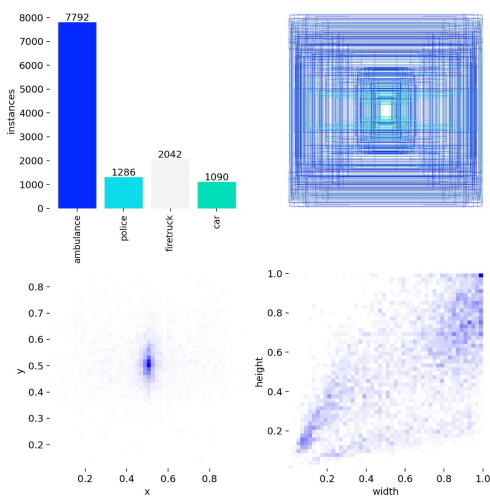
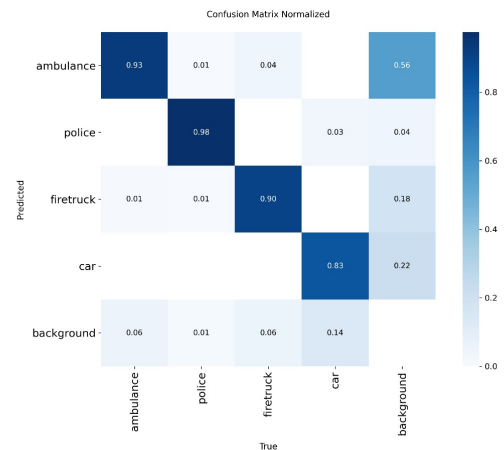
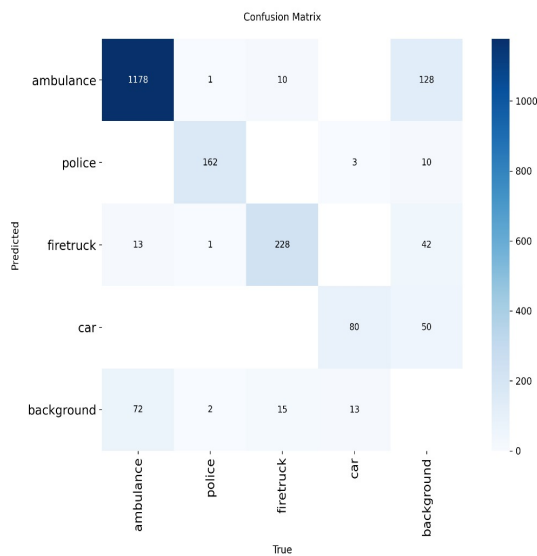
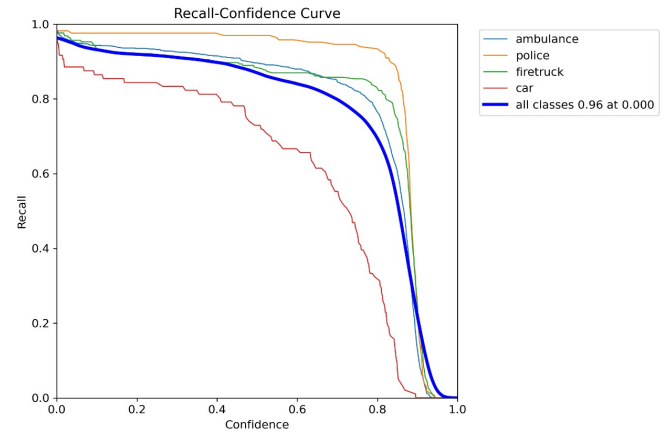
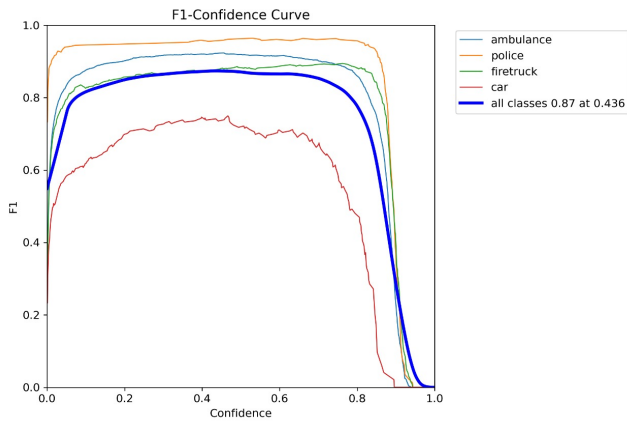


D. EV Prioritization Implementation

The main implementation of Emergency Vehicle Prioritization can be achieved through a client-server model, ensuring a continuous low latency feedback loop is provided. The main task involves the external AI client applying YOLOv8 for faster and more instantaneous detection of the position and pathway of the EV.

These results are then provided to the AI core, processing the optimal signal intervention with the shortest duration for intervention. The task of implementation is handled by TraCI, allowing for immediate signal intervention, including functions like "setPhase," within the SUMO simulation environment.





V. RESULTS, DISCUSSION, AND EVALUATION

The key element was the huge reduction in time to transport emergency cars. As the TraCI API gives a direct command, we could remove obstacles in real time, thus avoiding "zero-speed events" (waiting at red lights). This way, getting rid of these stopping situations, traveling speed will remain consistent. This is what one wants in an emergency, like a lifesaving case.

We also made sure that helping an ambulance didn't mean paralyzing the rest of the city. To handle this, the AI uses an "early release and late return" strategy. It waits until the very last safe second to flip the light to green and reverts to the normal schedule the moment the emergency vehicle clears the intersection. By measuring the average delay for regular drivers, we confirmed that this surgical

approach is much better for the city's overall flow than older systems that just "freeze" traffic for long, clunky intervals.

A. Technological Discussion and System Robustness

The discussion confirms the viability of the chosen technologies and the stability of the operational pipeline in a challenging, real-time control scenario.

1. "Called. On the Ground":

Running the numbers, it is evident that the best picks for this high-stakes task would be YOLOv8 models. It has performed very well, identifying the emergency vehicle every time correctly, and its speed is its main savior—it is processing the image within less than 10ms. Its speed is actually the only reason it managed to turn on the lights in time before the vehicle actually showed up; otherwise, the AI wouldn't have enough time to turn the lights around, which is exactly what one needs in an actual city scenario, and its latest "anchor-free" configuration is also not letting it get misled by the angles of the camera and distance of the vehicle, which is exactly the requisite here.

2. Smooth Operation: TraCI and SUMO In our case, the TraCI interaction type, which serves as the link between the SUMO context, functions in a highly efficient manner. There have been no delays or "noise" present in the interface between the logic AI and the traffic simulation environment. In fact, this serves as the very groundwork upon which our whole operation itself rests. This thus enables our AI "brain" to process many "emergency calls" at once and function effectively amidst the heavy traffic flow with nary a problem in sight. The fact that it can do extensive coding while transmitting signals automatically to the traffic lights with ease shows that this is much more than a lab project; in fact, it's fully functional in a real-world context.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

The integrated AI-Based Emergency Vehicle Prioritization System successfully validates a robust and effective methodology for real-time traffic signal intervention. By adopting the high-performance YOLOv8 model for low-latency, real-time vehicle detection and leveraging the SUMO-TraCI client-server architecture for instantaneous control, the project meets its primary objective: achieving an unimpeded "green wave" for emergency vehicles. The results confirmed a substantial, measurable reduction in EV travel duration and stops,

while the dynamic control strategy successfully minimized collateral disruption to general traffic flow. The modular, Python-integrated design also validates the ease of use and high reliability of the chosen open-source tools for complex, time-critical urban applications.

B. Future Applications

The AI-Based Emergency Vehicle Prioritization System, an integrated solution, successfully demonstrates that with the addition of YOLOv8, a fast detection approach, to SUMO-TraCI, a direct control approach, there can be an uninterrupted "green wave." This arrangement can drastically reduce emergency response times while ensuring normal traffic continues to flow as smoothly as possible. Looking ahead, very specifically, we are going to focus heavily on its robustness and usability with the integration of two major improvements: audio-visual fusion and a mobile app for emergency services staff to have a direct, real-time view of the status of this proven simulation, ensuring that this simulation can finally function as a useful, user-friendly operating solution for a life-saving endeavor. With the incorporation of an audio classification approach to identify sirens, this simulation will no longer merely see a situation; instead, it will hear a situation approaching, whether heavy rain or around a blind corner, drastically enhancing its reliability as a simulation solution to real-world situations.

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